

CPEN 416 / EECE 5710

Lecture 05: Multi-qubit states and gates

Thursday 18 September 2025

Announcements

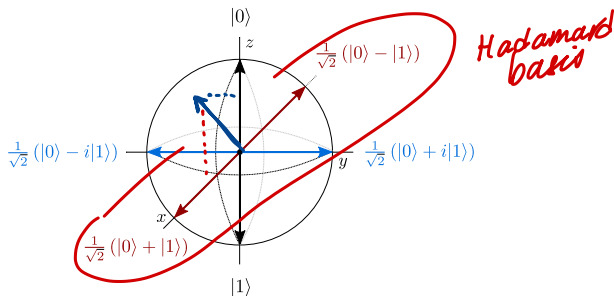
- Monday: tutorial assignment 1 (quantum key distribution)
- Quiz 3 on Tuesday
- Midterm in class on Thurs. 2 Oct (two weeks from today) - details and past midterms available ~~tomorrow~~ *now* on PrairieLearn

Last time

We formalized the idea of *projective measurements*: if we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$\Pr(\text{outcome } i) = |\langle\psi_i|\varphi\rangle|^2$$

Visually: geometric projection on the axis represented by a basis.



Last time

To measure in alternative bases, must apply a basis rotation.

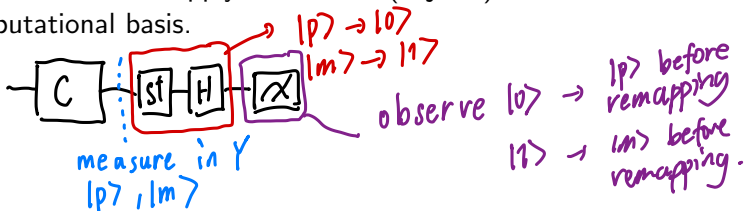
Example: measure in “Y” basis. First, determine mapping between computational basis and other basis:

$$|0\rangle \rightarrow |p\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

$$|1\rangle \rightarrow |m\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$$

$$\begin{aligned} |0\rangle &\xrightarrow{H} \xrightarrow{S} |p\rangle \\ |1\rangle &\xrightarrow{H} \xrightarrow{S} |m\rangle \end{aligned}$$

At the end of our circuit, apply the reverse (adjoint) to rotate *back* to the computational basis.



Last time: H - RZ exercise, revisited

Exercise: suppose we run the following circuit, but perform the measurement in the Y basis.



- 1 Using the Bloch sphere as intuition, what do you think the outcome probability of $|p\rangle$ looks like as a function of θ ? Discuss with a neighbour.
- 2 Test your predictions in software.

$$\Pr(|p\rangle)(\theta) :$$



Learning outcomes

- Mathematically describe a system of multiple qubits
- Describe the action of common multi-qubit gates
- Apply gates controlled on the state of multiple qubits
- Define and give examples of entangled states

Tensor products

Hilbert spaces compose under the *tensor product*.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

The tensor product of A and B , $A \otimes B$ is

$$A \otimes B = \begin{pmatrix} a \begin{pmatrix} e & f \\ g & h \end{pmatrix} & b \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ c \begin{pmatrix} e & f \\ g & h \end{pmatrix} & d \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ae & af & | & be & bf \\ ag & ah & | & bg & bh \\ \hline ce & cf & | & de & df \\ cg & ch & | & dg & dh \end{pmatrix}$$

$$C \otimes A \otimes B : 8 \times 8$$

$$\uparrow \\ 2 \times 2$$

$$D \otimes (C \otimes A \otimes B) : 16 \times 16$$

Multi-qubit states

Qubit state vectors also combine under the *tensor product*:

$$\begin{aligned}
 |0\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
 |1\rangle &= \begin{pmatrix} 0 \\ 1 \end{pmatrix}
 \end{aligned}
 \quad
 \underbrace{|0\rangle \otimes |0\rangle}_{|00\rangle} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Exercise: Determine the 2-qubit computational basis.

$$\begin{aligned}
 &^0 |00\rangle, \quad ^1 |01\rangle, \quad ^2 |10\rangle, \quad ^3 |11\rangle \\
 &\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}
 \end{aligned}$$

$$2^n \quad \underbrace{11}$$

Question: How many n -qubit basis states? (dimension of space?)

Multi-qubit states

The tensor product is linear and distributive. Given

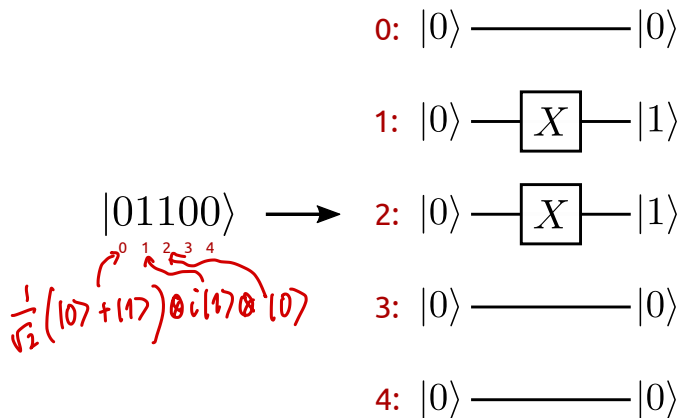
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

$$\begin{aligned}
 |\psi\rangle \otimes |\varphi\rangle &= (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) \\
 &= \alpha\gamma|0\rangle \otimes |0\rangle + \alpha\delta|0\rangle \otimes |1\rangle + \beta\gamma|1\rangle \otimes |0\rangle + \beta\delta|1\rangle \otimes |1\rangle \\
 &= \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} \rightarrow |\alpha\gamma|^2 + |\alpha\delta|^2 + |\beta\gamma|^2 + |\beta\delta|^2 = 1 \\
 &\quad |\alpha|^2(|\gamma|^2 + |\delta|^2) + \dots \\
 &\quad |\alpha\gamma|^2 = (\alpha\gamma)(\alpha^*\gamma^*)
 \end{aligned}$$

Question: Are tensor products of states still normalized? ✓

Qubit ordering (very important!)

In PennyLane:



This is different in other frameworks.

Multi-qubit gates

★ TODO: tutorial questions

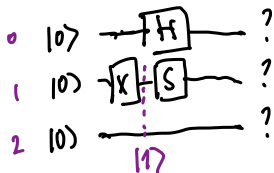
Unitary operations also compose under tensor product.

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes i|1\rangle \otimes |0\rangle$$

Example: apply U_1 to $|\psi\rangle$ and U_2 to $|\varphi\rangle$

$$(U_1 \otimes U_2) |\psi\rangle \otimes |\varphi\rangle = (U_1 |\psi\rangle) \otimes (U_2 |\varphi\rangle)$$

$|\psi\varphi\rangle$

Exercise: determine the state of a 3-qubit system if H is applied to qubit 0, X and then S are applied to qubit 1.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} i|1\rangle \\ |0\rangle \end{pmatrix}$$

$$i|10\rangle = \begin{pmatrix} 0 \\ 0 \\ i \\ 0 \end{pmatrix}$$

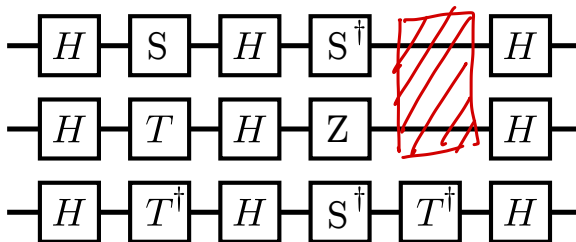
$$\frac{1}{\sqrt{2}} i (|010\rangle + |110\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} i \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \\ i \\ 1 \end{pmatrix}$$

$$i|1+10\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \\ i \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \\ i \\ 1 \end{pmatrix}$$

Multi-qubit gates

The circuits we've seen so far only involve single-qubit gates:



Surely this isn't all we can do...

Image credit: Xanadu Quantum Codebook I.11

SWAP

Define an operation that swaps the state of two qubits. First, specify action on basis states:

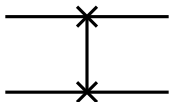
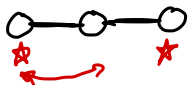
$$\begin{aligned}
 |00\rangle &\rightarrow |00\rangle \star \\
 |01\rangle &\rightarrow |10\rangle \\
 |10\rangle &\rightarrow |01\rangle \\
 |11\rangle &\rightarrow |11\rangle \star
 \end{aligned}$$

$SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

(Note: The matrix has a red dashed box around the $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ sub-block, and a red arrow points from this box to the mapping $|01\rangle \rightarrow |10\rangle$ and $|10\rangle \rightarrow |01\rangle$. Blue arrows indicate the mapping of $|00\rangle$ to $|00\rangle$ and $|11\rangle$ to $|11\rangle$. A blue arrow points to the bottom-right element '1' in the matrix.)

• four 1s

Circuit element:



$$\begin{aligned}
 &\left. \begin{array}{l} |0\rangle \\ |0\rangle \end{array} \right\} \xrightarrow{\text{H}} \left. \begin{array}{l} |0\rangle \\ |1\rangle \end{array} \right\} \\
 &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{H}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\
 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}
 \end{aligned}$$

PennyLane: `qml.SWAP`

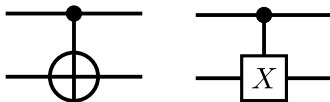
CNOT

CNOT = “controlled-NOT”. A NOT (X) is applied to second qubit only if first qubit is in state $|1\rangle$.

$|00\rangle \rightarrow |00\rangle$
 $|01\rangle \rightarrow |01\rangle$
 $|10\rangle \rightarrow |11\rangle$
 $|11\rangle \rightarrow |10\rangle$

$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

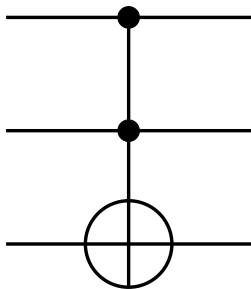
Circuit elements:



PennyLane: `qml.CNOT`

Toffoli (TOF)

There are also gates on more than two qubits, like the Toffoli gate, which is a controlled-CNOT, or controlled-controlled-NOT.



$$TOF|000\rangle = \rightarrow$$

$$TOF|001\rangle = \rightarrow$$

$$TOF|010\rangle =$$

$$TOF|011\rangle =$$

$$TOF|100\rangle =$$

$$TOF|101\rangle =$$

$$TOF|110\rangle = |111\rangle$$

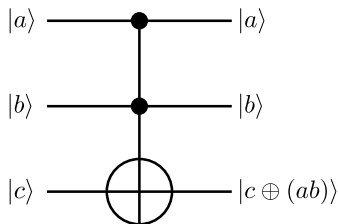
$$TOF|111\rangle = |110\rangle$$

PennyLane: `qml.Toffoli`

Quantum circuits and Boolean functions

The Toffoli implements a reversible AND gate. It is also universal for classical reversible computing.

a	b	$\wedge a.b$
0	0	0
0	1	0
1	0	0
1	1	1



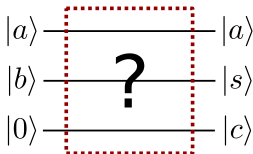
Exercise: what Boolean function does CNOT implement?



Quantum circuits and Boolean functions

X, CNOT, TOF can be used to create Boolean arithmetic circuits.

Exercise: use a combination of these gates to create a quantum version of a one-bit adder. Test it in PennyLane on all possible bit values of a and b .



Do as homework - will discuss
on Tuesday.

Next time

Content:

qml. ctrl
✓ Entanglement

- Multi-qubit measurement and the Bell basis
- Superdense coding

Action items:

- ① Work on practice assignment 1 (can do Qs 1-4)
- ② Work through last section of Monday's tutorial (bases)

Recommended reading:

- Codebook nodes IQC, SQ, MQ; Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3